

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Two carts, A and B, are on a horizontal, frictionless track.

- Cart A: mass = 0.30 kg, moving at 0.80 m/s
- Cart B: mass = 0.50 kg, at rest
- After collision: Cart A moves left at 0.20 m/s

(a)

Calculate the velocity of cart B after the collision.

Use conservation of momentum:

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

Substitute values:

$$0.30 \times 0.80 + 0.50 \times 0 = 0.30 \times (-0.20) + 0.50 \times v_B$$

$$0.24 = -0.06 + 0.50v_B \Rightarrow 0.30 = 0.50v_B \Rightarrow v_B = 0.60 \text{ m/s (to the right)}$$

 *Revision tip:*
Define **one direction as positive** (e.g. right).
Use negative for motion in the opposite direction.

(b)

Calculate the total kinetic energy before and after the collision. Does this indicate an elastic or inelastic collision?

Before:

Only cart A has KE:

$$E_{k\text{before}} = \frac{1}{2} \times 0.30 \times 0.80^2 = 0.096 \text{ J}$$

After:

Cart A:

$$E_{kA} = \frac{1}{2} \times 0.30 \times 0.20^2 = 0.006 \text{ J}$$

Cart B:

$$E_{kB} = \frac{1}{2} \times 0.50 \times 0.60^2 = 0.090 \text{ J}$$

$$E_{k\text{after}} = 0.006 + 0.090 = 0.096 \text{ J}$$

 Total KE is **conserved** → this is an **elastic collision**.

 *Revision tip:*
Elastic collision = **momentum and kinetic energy conserved**
Inelastic = **momentum conserved, KE not**